Modelling the Variance: ARCH / GARCH Models

Econometrics. Lecture 11

Content

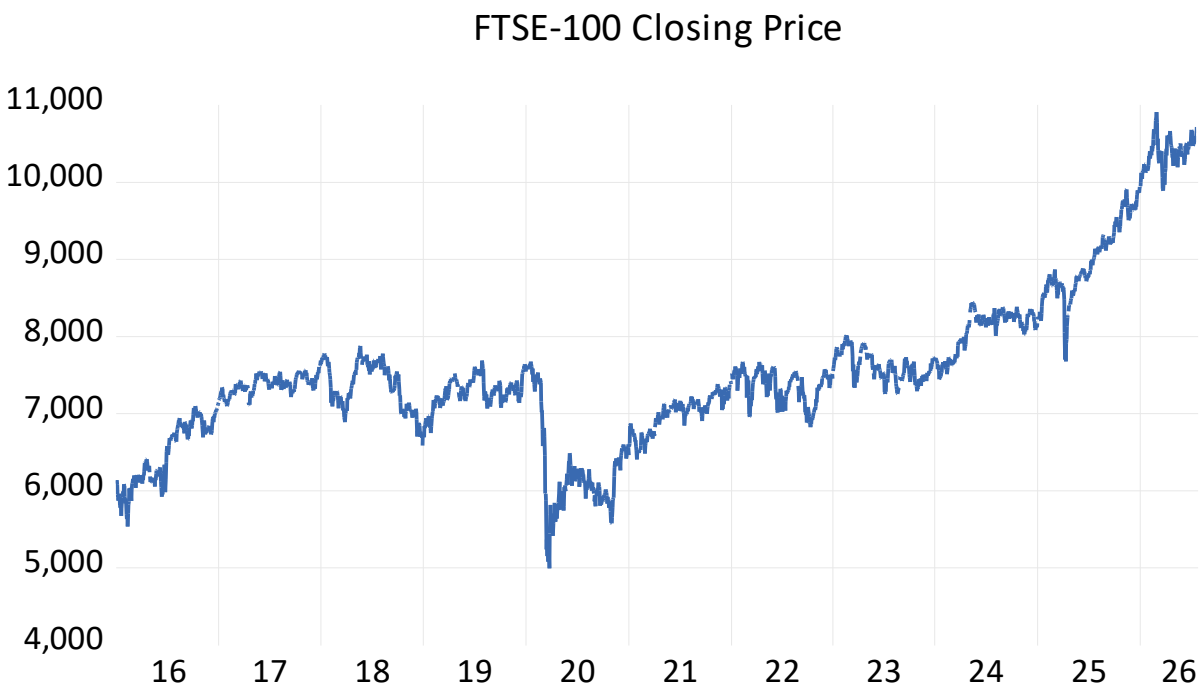
1. Volatility clustering
2. ARCH model
3. GARCH model
4. EGARCH model

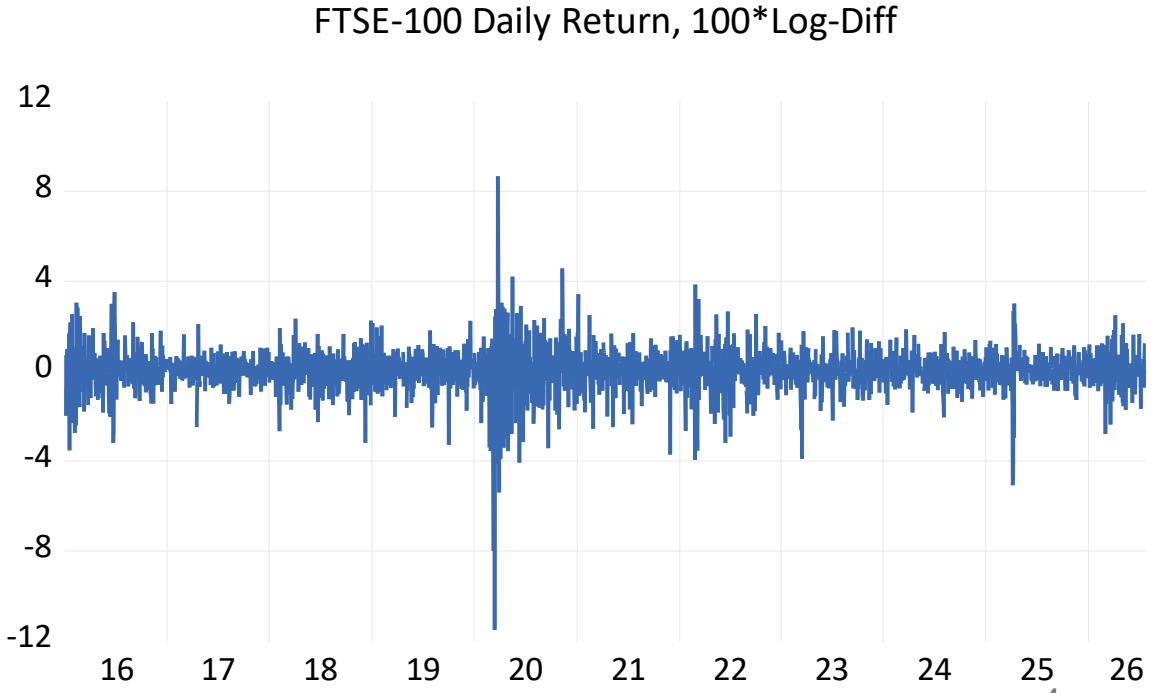
1. Volatility clustering

Volatility clustering

The volatility of many financial and macroeconomic variables changes over time.

A series with some periods of low volatility and some periods of high volatility is said to exhibit **volatility clustering.**





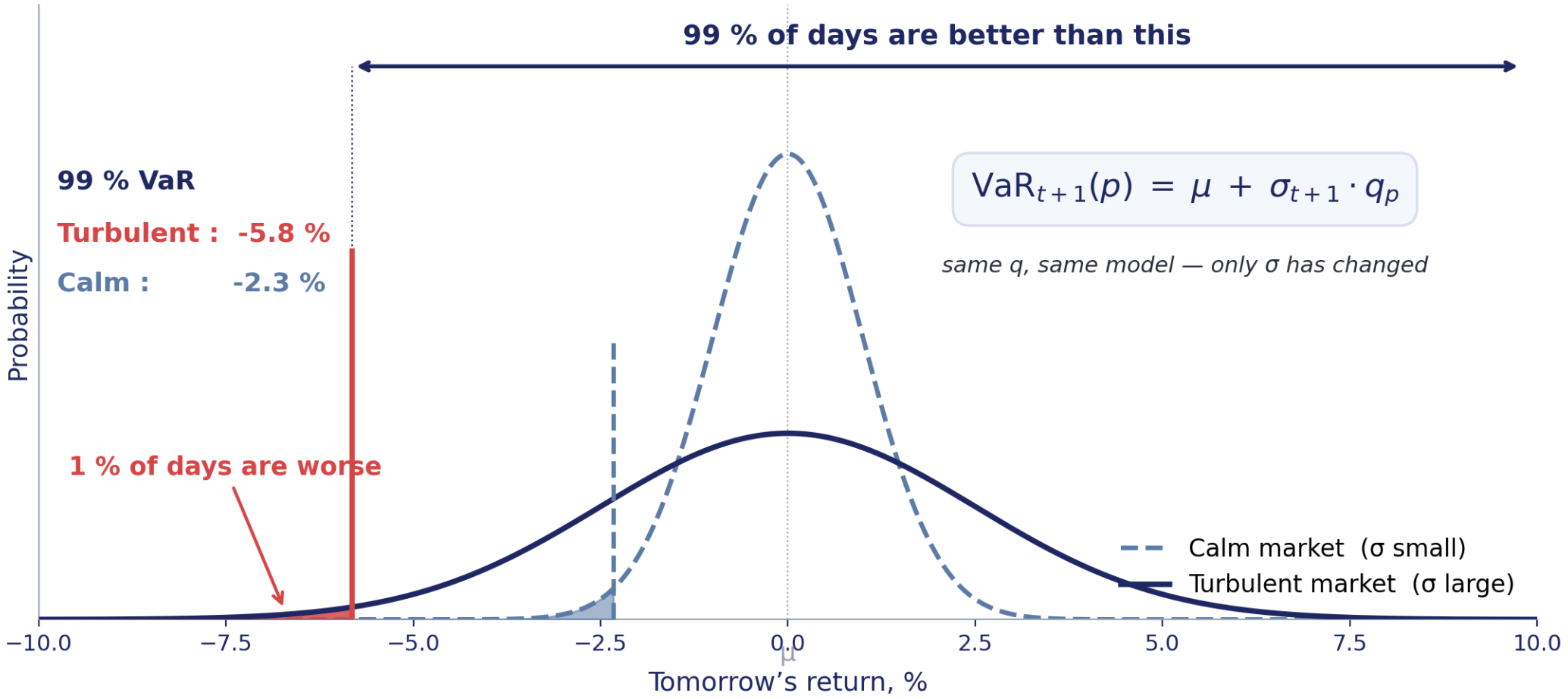
Why forecasting the variance?

1. To measure the risk of owning an asset.

2. To know the price at which to buy or sell some financial derivates.

3. To improve the accuracy of forecast intervals.

Why forecasting the variance?
The risk limit moves with it



2. ARCH Model

ARCH(1)

Engle, R. (1982). Autoregressive Conditional Heteroskedasticity with Estimates of the Variance of U.K. Inflation. *Econometrica*, 50(4), 987-1007.

ARCH(1) model will simultaneously model the mean and the variance of the series:

Mean equation: $y_t = \beta_0 + \beta_1 y_{t-1} + \beta_2 x_t + u_t,$ where $u_t|\Omega_t \sim iid\ N(0, \sigma_t^2),$

where Ω_t is the information set

Variance equation: $\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2,$

Covariance-stationarity condition $\longrightarrow$ $\alpha_0 > 0,$ $0 \leq \alpha_1 < 1$

Unconditional variance is $Var(u_t) = \alpha_0/(1 - \alpha_1)$

ARCH(q)

ARCH(q) process:

Mean equation: $y_t = \beta_0 + \beta_1 y_{t-1} + \beta_2 x_t + u_t$

Variance equation: $\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \alpha_2 u_{t-2}^2 + \cdots + \alpha_q u_{t-q}^2 = \alpha_0 + \sum_{j=1}^{q} \alpha_j u_{t-j}^2$

$\alpha_0 > 0, \ \ \alpha_j \geq 0$ for $j = 1, \ldots, q.$

Covariance-stationarity condition $\longrightarrow$ $\sum_{j=1}^{q} \alpha_j < 1$

Unconditional variance is $Var(u_t) = \alpha_0 / (1 - \sum \alpha_j)$

Testing for ARCH effects.
ARCH LM Test

Let's assume that we have the following variance equation:

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \alpha_2 u_{t-2}^2 + \cdots + \alpha_q u_{t-q}^2$$

The **ARCH Lagrange multiplier (LM) test** involves the following steps:

1) Obtain the residuals $\hat{u}_t$ from your "best" regression model.
2) Run the auxiliary regression of squared residuals $\hat{u}_i^2$:

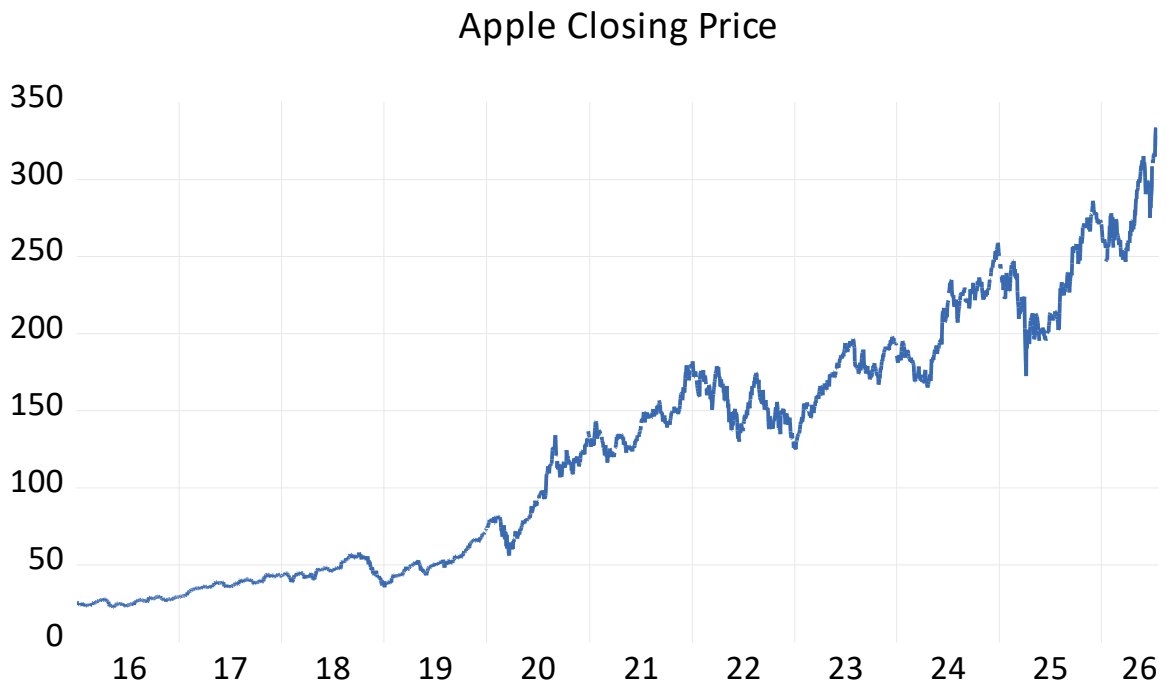
$$\hat{u}_t^2 = \delta_0 + \delta_1 u_{t-1}^2 + \delta_2 u_{t-2}^2 + \cdots + \delta_q u_{t-q}^2 + v$$

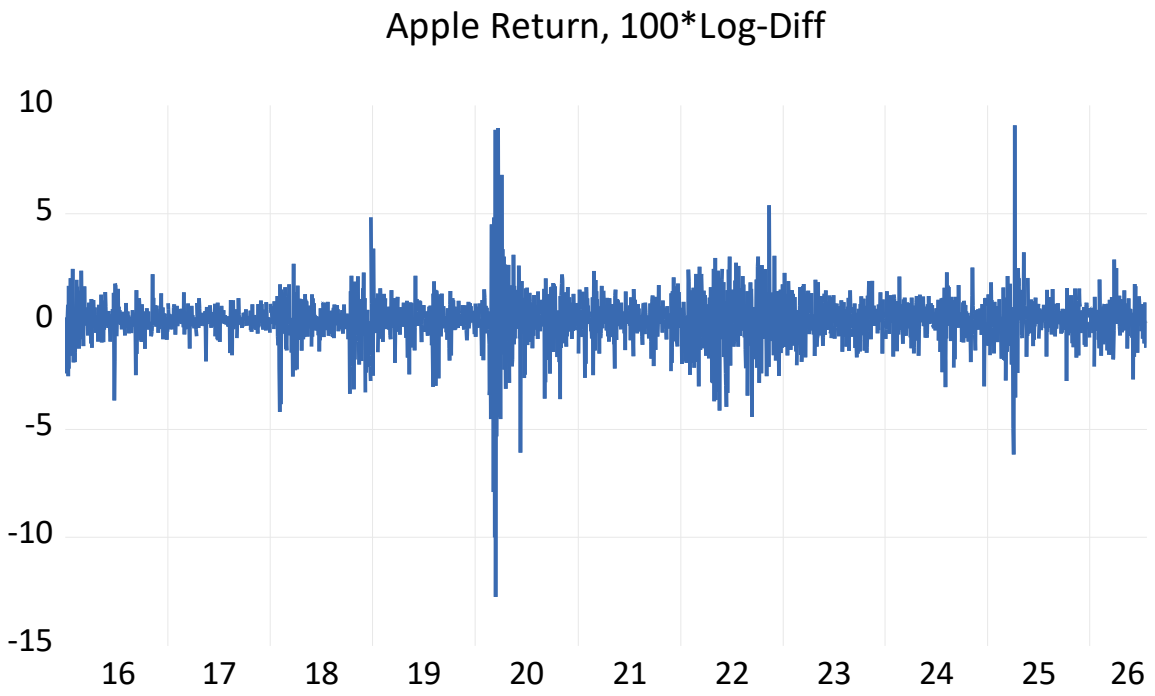
3) Formulate the null and the alternative hypotheses:

$$\mathrm{H}_0\colon\ \delta_1 = \delta_2 = \cdots = \delta_q = 0 \quad \text{vs} \quad \mathrm{H}_1\colon at\ least\ one\ of\ \delta_1, \delta_2, \ldots, \delta_q\ is\ different\ from\ 0$$

4) Compute the $LM = nR^2$ statistic, where *n* and R^2 is the number of observations and *R*-squared in the auxiliary regression from step 2 respectively. The *LM*-statistic follows the χ_q^2 distribution.
5) If the *p*-value of *LM*-statistic is sufficiently small, that is, below the chosen significance level, then we reject the null hypothesis of no ARCH effects.

EViews example: Apple stock price. Visual inspection of the time series





EViews example.
Estimate the mean equation

Dependent Variable: DLOG(PRICE)*100
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 09/09/26 Time: 09:47
Sample: 1/05/2016 7/24/2026
Included observations: 2653
Convergence achieved after 31 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.095621	0.033907	2.820079	0.0048
MA(1)	-0.053520	0.011870	-4.508933	0.0000
SIGMASQ	3.305594	0.047751	69.22616	0.0000

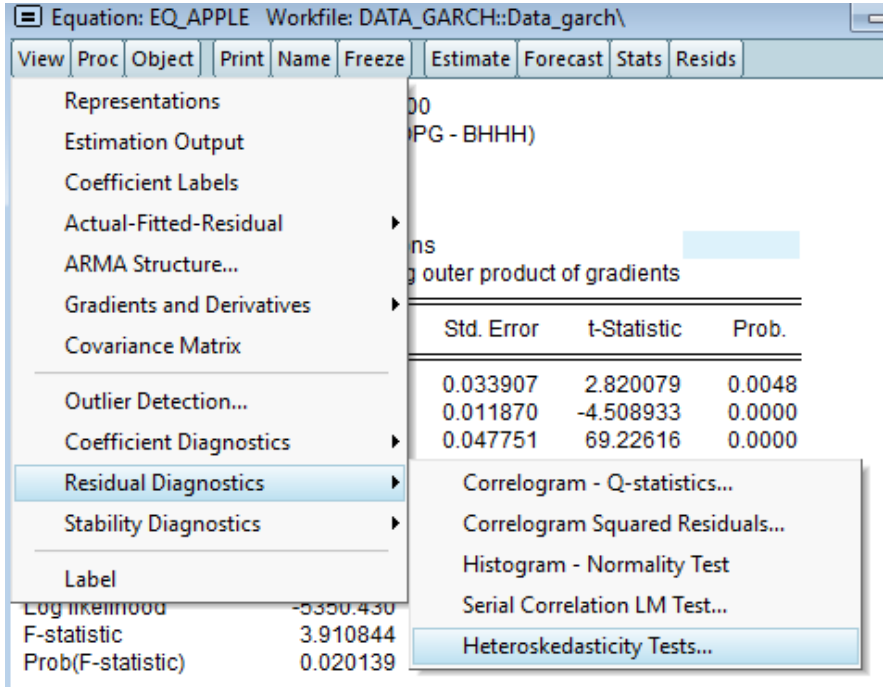
R-squared	0.002943	Mean dependent var	0.095632
Adjusted R-squared	0.002190	S.D. dependent var	1.821154
S.E. of regression	1.819158	Akaike info criterion	4.035756
Sum squared resid	8769.741	Schwarz criterion	4.042409
Log likelihood	-5350.430	Hannan-Quinn criter.	4.038164
F-statistic	3.910844	Durbin-Watson stat	1.999623
Prob(F-statistic)	0.020139		

Inverted MA Roots	.05

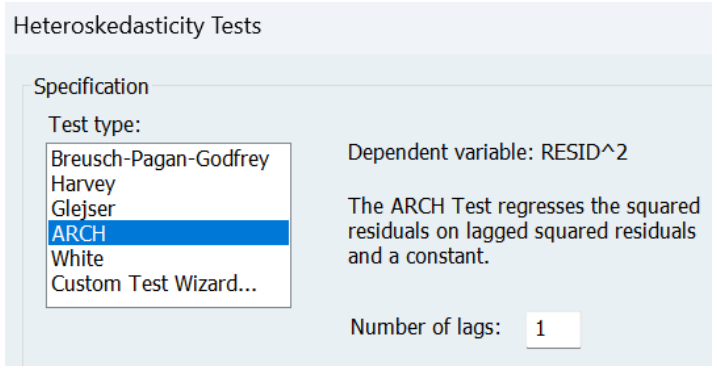
By default (when estimating ARCH/GARCH later), work in percent returns (×100). Daily returns near 0.001 make the likelihood badly scaled for the optimizer; rescaling to order-1 improves convergence and standard errors without changing the model.

EViews example.
Testing for ARCH effects

Testing for ARCH effects



Test at lag 1 as the baseline



Heteroskedasticity Test: ARCH

F-statistic	189.9461	Prob. F(1,2650)	0.0000
Obs*R-squared	177.3756	Prob. Chi-Square(1)	0.0000

Test Equation:
Dependent Variable: RESID^2
Method: Least Squares
Date: 09/09/26 Time: 09:48
Sample (adjusted): 1/06/2016 7/24/2026
Included observations: 2652 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	2.450003	0.187678	13.05427	0.0000
RESID^2(-1)	0.258643	0.018767	13.78209	0.0000

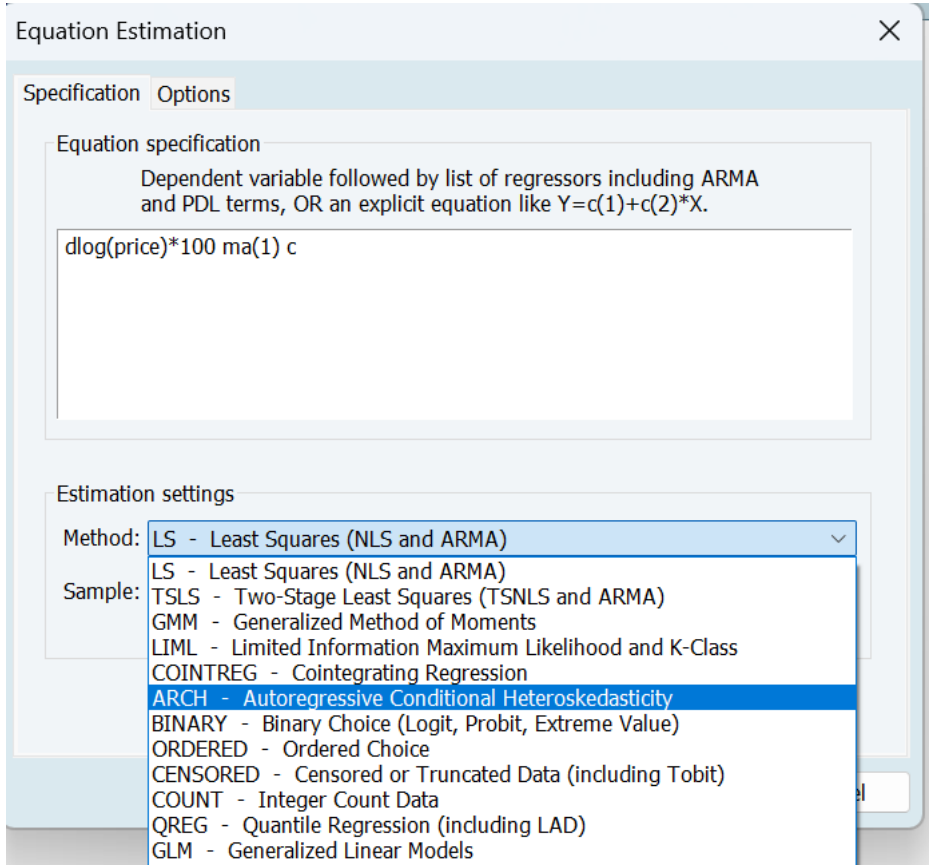
R-squared	0.066884	Mean dependent var	3.304233
Adjusted R-squared	0.066532	S.D. dependent var	9.442209
S.E. of regression	9.122701	Akaike info criterion	7.260163
Sum squared resid	220542.7	Schwarz criterion	7.264599
Log likelihood	-9624.976	Hannan-Quinn criter.	7.261769
F-statistic	189.9461	Durbin-Watson stat	2.086963
Prob(F-statistic)	0.000000		

LM-statistic

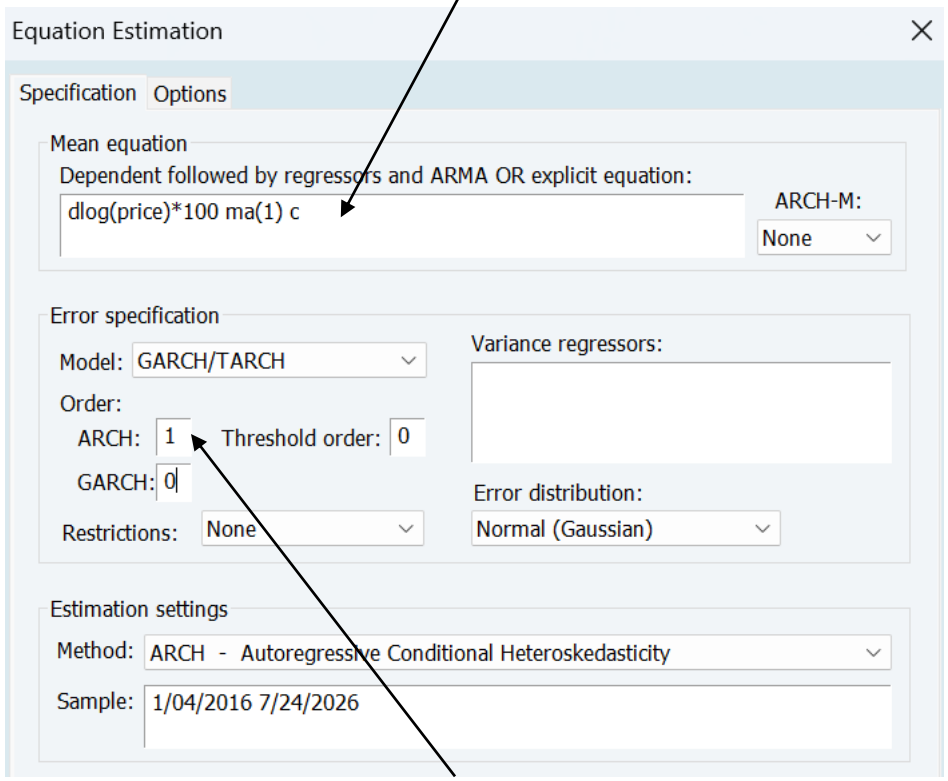
p-value = 0.000 < 0.05

We reject H_0: *no ARCH effects*.

EViews example.
Estimate ARCH model (1)



Mean equation specification



The number of ARCH components (we estimate a simple ARCH(1) model – one lag of the squared shocks in the variance equation.

EViews example.
Estimate ARCH model (2)

Dependent Variable: DLOG(PRICE)*100
Method: ML ARCH - Normal distribution (OPG - BHHH / Marquardt steps)
Date: 09/09/26 Time: 09:52
Sample (adjusted): 1/05/2016 7/24/2026
Included observations: 2653 after adjustments
Convergence not achieved after 500 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1/04/2016
Presample variance: backcast (parameter = 0.7)
GARCH = C(3) + C(4)*RESID(-1)^2

Variable	Coefficient	Std. Error	z-Statistic	Prob.
C	0.132397	0.026805	4.939330	0.0000
MA(1)	-0.030431	0.014069	-2.163035	0.0305
Variance Equation				
C	2.178659	0.035501	61.36844	0.0000
RESID(-1)^2	0.171429	0.011652	14.71228	0.0000

R-squared	0.001962	Mean dependent var	0.095632
Adjusted R-squared	0.001586	S.D. dependent var	1.821154
S.E. of regression	1.819709	Akaike info criterion	3.968196
Sum squared resid	8778.368	Schwarz criterion	3.977067
Log likelihood	-5259.813	Hannan-Quinn criter.	3.971407
Durbin-Watson stat	2.045536		

Inverted MA Roots .03

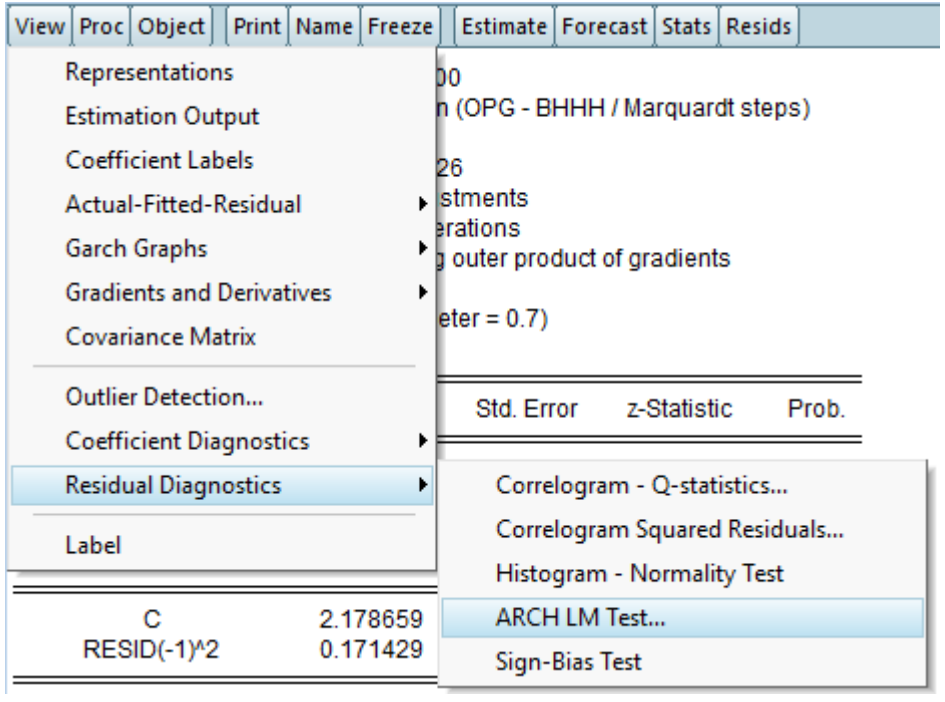
Positive as required

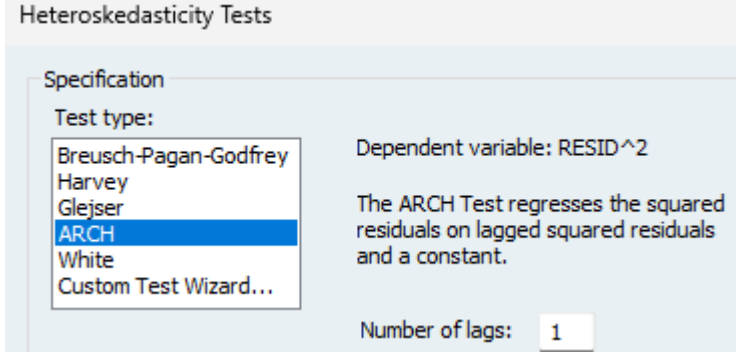
Not lower than 0 and lower than 1 as required.

Significant at 1% level.

The conditional variance is stationary.

EViews example.
Testing for ARCH(1) effects





Heteroskedasticity Test: ARCH

F-statistic	0.126931	Prob. F(1,2650)	0.7217
Obs*R-squared	0.127020	Prob. Chi-Square(1)	0.7215

Test Equation:
Dependent Variable: WGT_RESID^2
Method: Least Squares
Date: 09/09/26 Time: 09:53
Sample (adjusted): 1/06/2016 7/24/2026
Included observations: 2652 after adjustments

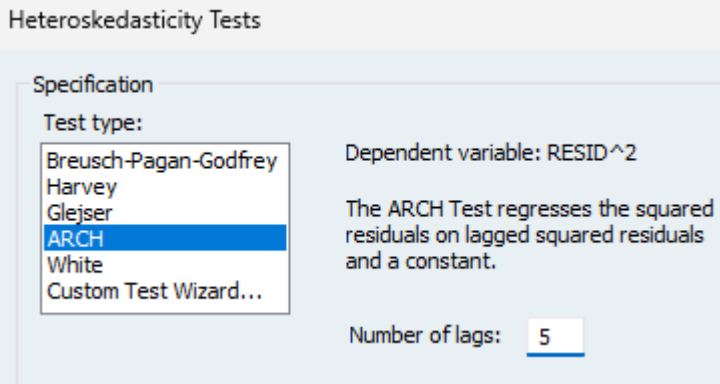
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.186151	0.061958	19.14443	0.0000
WGT_RESID^2(-1)	-0.006922	0.019429	-0.356273	0.7217

R-squared	0.000048	Mean dependent var	1.178002
Adjusted R-squared	-0.000329	S.D. dependent var	2.964847
S.E. of regression	2.965336	Akaike info criterion	5.012611
Sum squared resid	23302.02	Schwarz criterion	5.017048
Log likelihood	-6644.723	Hannan-Quinn criter.	5.014217
F-statistic	0.126931	Durbin-Watson stat	1.998004
Prob(F-statistic)	0.721664		

We fail to reject H_0: $no\ remaining\ ARCH(1)\ effect$.

EViews example.
Higher-order ARCH model (1)

Test at lag 5 (one trading week) from the previous model



LM-statistic

p-value = 0.000 < 0.05

We reject H_0: $no\ ARCH\ effects$.

Heteroskedasticity Test: ARCH

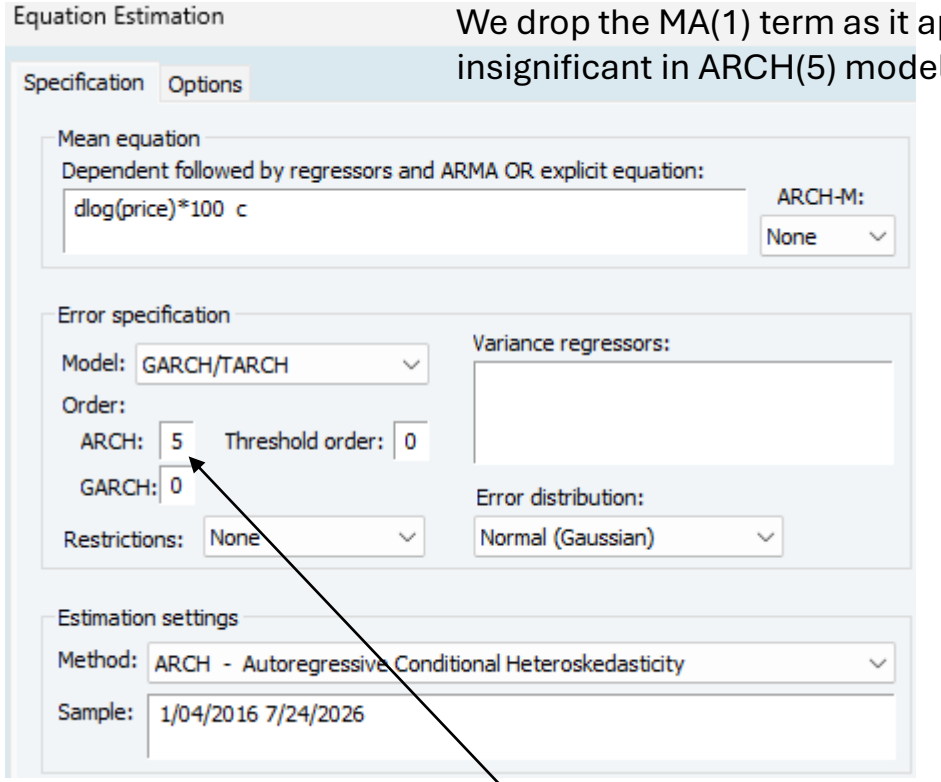
F-statistic	24.59107	Prob. F(5,2642)	0.0000
Obs*R-squared	117.7544	Prob. Chi-Square(5)	0.0000

Test Equation:
Dependent Variable: WGT_RESID^2
Method: Least Squares
Date: 09/09/26 Time: 09:54
Sample (adjusted): 1/12/2016 7/24/2026
Included observations: 2648 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.757182	0.072091	10.50318	0.0000
WGT_RESID^2(-1)	-0.037040	0.019325	-1.916682	0.0554
WGT_RESID^2(-2)	0.094262	0.019238	4.899859	0.0000
WGT_RESID^2(-3)	0.080526	0.019249	4.183341	0.0000
WGT_RESID^2(-4)	0.101265	0.019226	5.267110	0.0000
WGT_RESID^2(-5)	0.117083	0.019314	6.062207	0.0000

R-squared	0.044469	Mean dependent var	1.176324
Adjusted R-squared	0.042661	S.D. dependent var	2.965047
S.E. of regression	2.901112	Akaike info criterion	4.970328
Sum squared resid	22236.26	Schwarz criterion	4.983655
Log likelihood	-6574.715	Hannan-Quinn criter.	4.975152
F-statistic	24.59107	Durbin-Watson stat	2.017912
Prob(F-statistic)	0.000000		

EViews example.
Higher-order ARCH model (2)



We drop the MA(1) term as it appears insignificant in ARCH(5) model

We include 5 ARCH components (five lags of the squared shocks)

Positive as required

Each coefficient is not lower than zero as required and is significant at 1% level.

Their sum is 0.573 – lower than 1 as required.

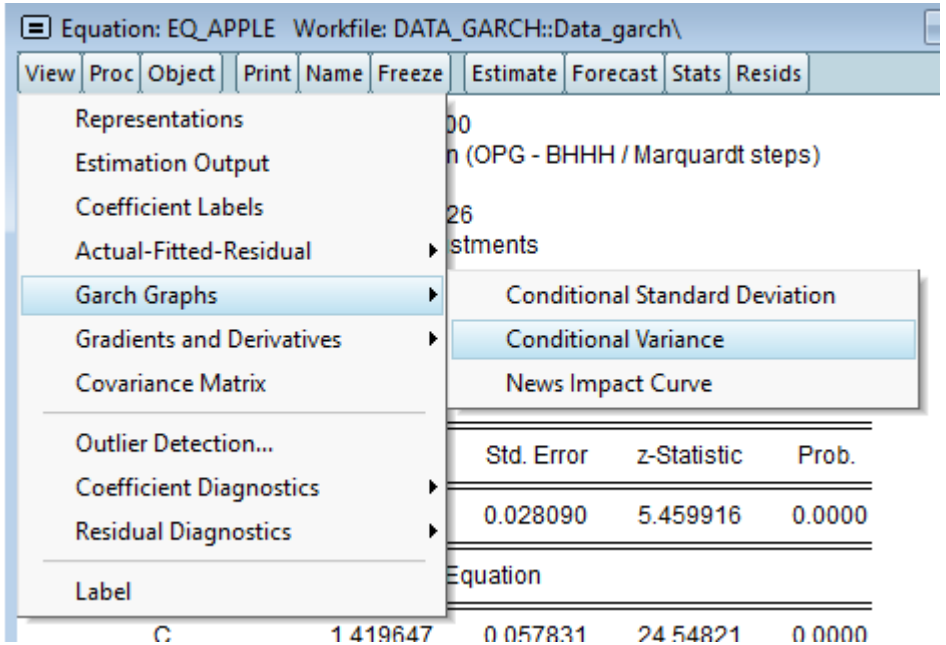
Our conditional variance is stationary.

Dependent Variable: DLOG(PRICE)*100
Method: ML ARCH - Normal distribution (OPG - BHHH / Marquardt steps)
Date: 09/09/26 Time: 09:56
Sample (adjusted): 1/05/2016 7/24/2026
Included observations: 2653 after adjustments
Convergence achieved after 40 iterations
Coefficient covariance computed using outer product of gradients
Presample variance: backcast (parameter = 0.7)
GARCH = C(2) + C(3)*RESID(-1)^2 + C(4)*RESID(-2)^2 + C(5)*RESID(-3)^2 + C(6)*RESID(-4)^2 + C(7)*RESID(-5)^2

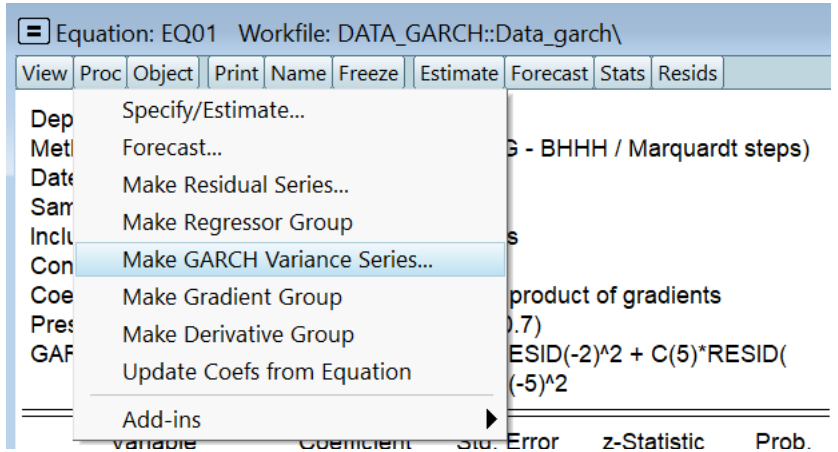
Variable	Coefficient	Std. Error	z-Statistic	Prob.
C	0.153369	0.028090	5.459916	0.0000
Variance Equation				
C	1.419647	0.057831	24.54821	0.0000
RESID(-1)^2	0.138020	0.021762	6.342318	0.0000
RESID(-2)^2	0.093084	0.018375	5.065741	0.0000
RESID(-3)^2	0.106451	0.016857	6.314876	0.0000
RESID(-4)^2	0.105410	0.015506	6.798112	0.0000
RESID(-5)^2	0.129971	0.015642	8.309139	0.0000

R-squared	-0.001005	Mean dependent var	0.095632
Adjusted R-squared	-0.001005	S.D. dependent var	1.821154
S.E. of regression	1.822069	Akaike info criterion	3.856353
Sum squared resid	8804.469	Schwarz criterion	3.871876
Log likelihood	-5108.452	Hannan-Quinn criter.	3.861972
Durbin-Watson stat	2.105902		

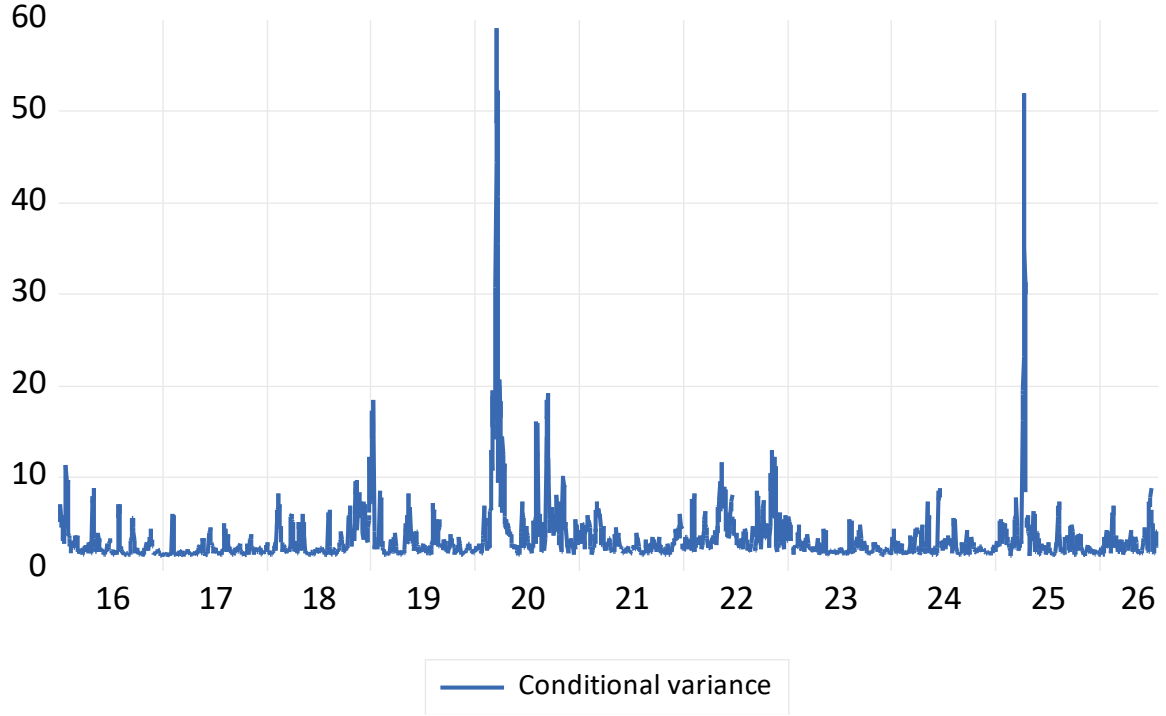
EViews example.
Obtaining the conditional volatility (variance)

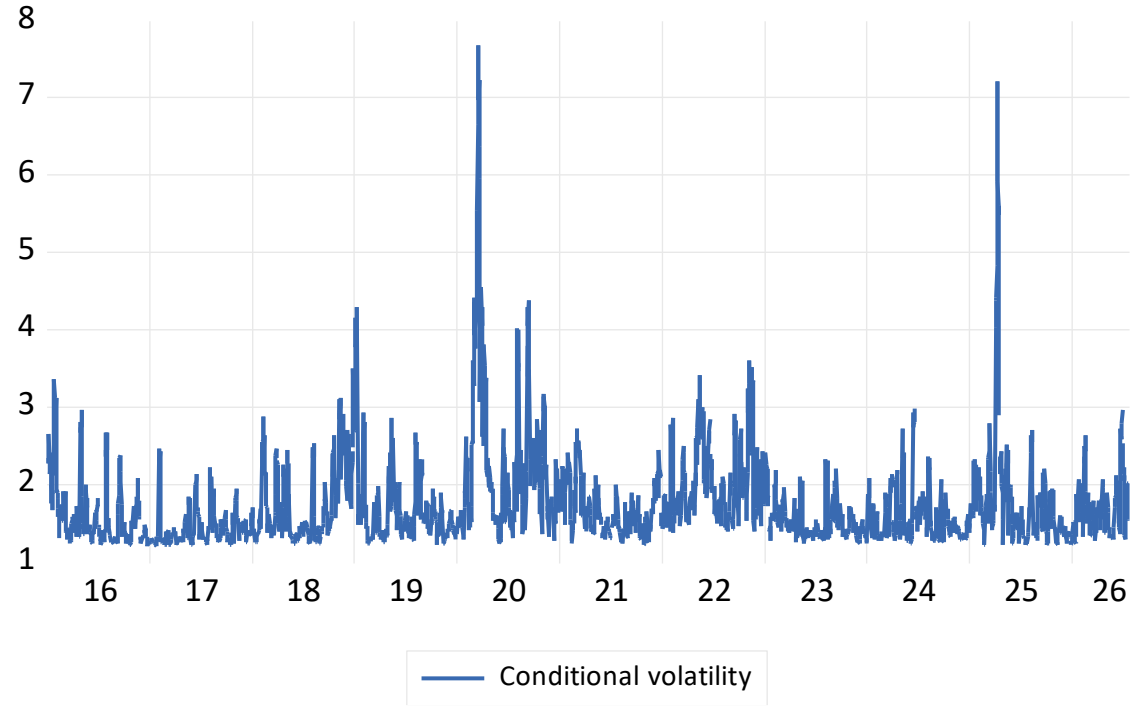


or save the GARCH series in the workfile



EViews example.
Conditional variance and volatility





Conditional volatility is the ARCH(5) model's daily estimate of the stock's risk – the expected size of the daily return move, from about ±1.7% on calm days to near ±7% at the peaks.

Its key message is that risk is not constant but clusters in time: shocks push volatility up sharply, after which it decays only slowly back toward its long-run average.

Drawbacks of the ARCH specification

- ARCH needs too many lags to capture volatility persistence. It looked more like a moving average specification than an autoregression.

- A new idea was born which was to include the lagged conditional variance terms as autoregressive terms.

- This idea was worked out by Tim Bollerslev, who in 1986 published a paper entitled “Generalised Autoregressive Conditional Heteroskedasticity” in *Journal of Econometrics*.

3. GARCH Model

GARCH(1,1)

The GARCH model extends the ARCH model to let σ_t^2 depend on its own lags as well as lags of the squared error:

Mean equation: $y_t = \beta_0 + \beta_1 y_{t-1} + \beta_2 x_t + u_t,$ where $u_t|\Omega_t \sim iid\ N(0, \sigma_t^2),$

where Ω_t is the information set

Variance equation: $\sigma_t^2 = \alpha_0 + \phi_1 \sigma_{t-1}^2 + \alpha_1 u_{t-1}^2,$

$$\alpha_0 > 0,\ \alpha_1 \geq 0,\ \phi_1 \geq 0$$

Covariance-stationarity condition $\longrightarrow$ $\alpha_1 + \phi_1 < 1$

Unconditional variance is $Var(u_t) = \alpha_0/(1 - \alpha_1 - \phi_1)$

GARCH(*p*,*q*)

The GARCH model extends the ARCH model to let σ_t^2 depend on its own lags as well as lags of the squared error:

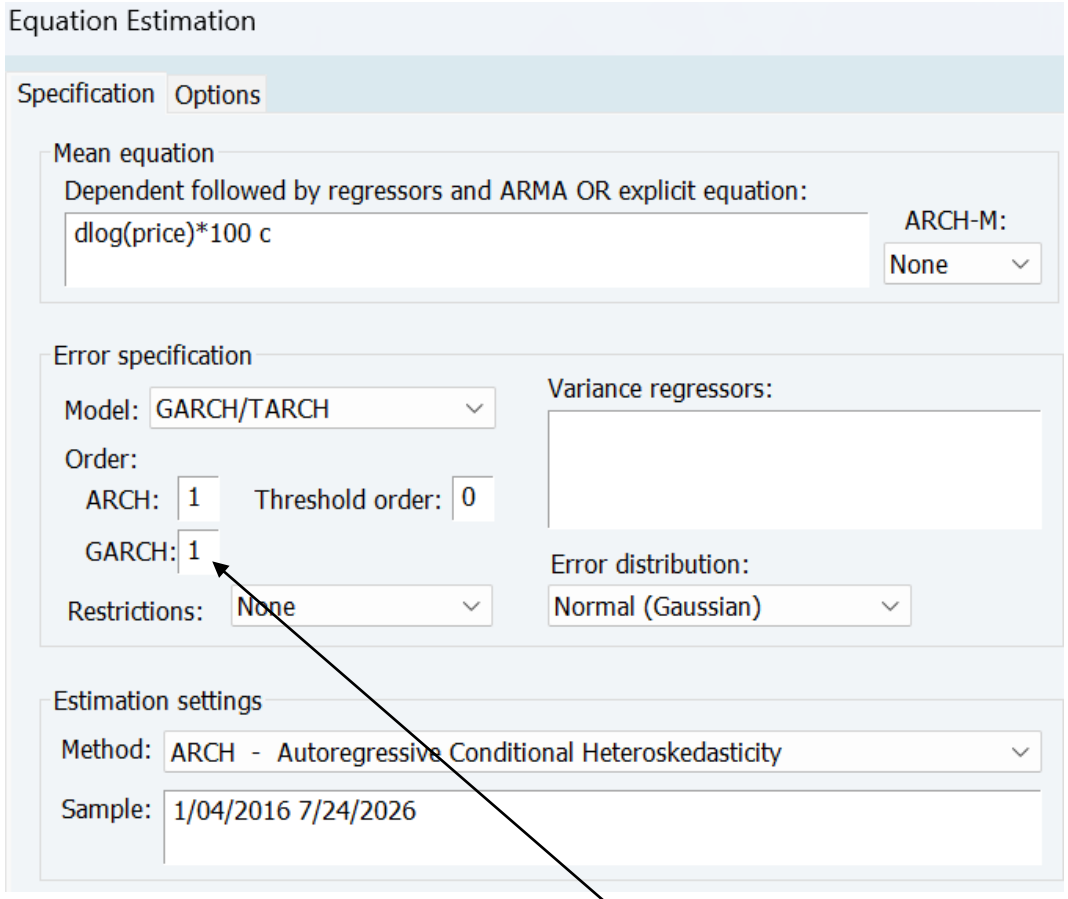
Mean equation: $y_t = \beta_0 + \beta_1 y_{t-1} + \beta_2 x_t + u_t$

Variance equation: $\sigma_t^2 = \alpha_0 + \phi_1 \sigma_{t-1}^2 + \cdots + \phi_p \sigma_{t-p}^2 + \alpha_1 u_{t-1}^2 + \cdots + \alpha_q u_{t-q}^2 = \alpha_0 + \sum_{i=1}^{p} \phi_i \, \sigma_{t-i}^2 + \sum_{j=1}^{q} \alpha_j \, u_{t-j}^2$

$\alpha_0 > 0; \phi_i \geq 0$ for $i = 1, \ldots, p;$ $\alpha_j \geq 0$ for $j = 1, \ldots, q.$

Covariance-stationarity condition $\rightarrow \sum_{i=1}^{p} \phi_i + \sum_{j=1}^{q} \alpha_j < 1$

EViews example. GARCH(1,1)



We include the lag of the conditional variance

Dependent Variable: DLOG(PRICE)*100
Method: ML ARCH - Normal distribution (OPG - BHHH / Marquardt steps)
Date: 09/09/26 Time: 10:31
Sample (adjusted): 1/05/2016 7/24/2026
Included observations: 2653 after adjustments
Convergence achieved after 38 iterations
Coefficient covariance computed using outer product of gradients
Presample variance: backcast (parameter = 0.7)
GARCH = C(2) + C(3)*RESID(-1)^2 + C(4)*GARCH(-1)

Variable	Coefficient	Std. Error	z-Statistic	Prob.
C	0.157004	0.028397	5.528819	0.0000
Variance Equation				
C	0.142350	0.016576	8.587796	0.0000
RESID(-1)^2	0.098731	0.009773	10.10281	0.0000
GARCH(-1)	0.856238	0.012623	67.83337	0.0000

R-squared	-0.001136	Mean dependent var	0.095632
Adjusted R-squared	-0.001136	S.D. dependent var	1.821154
S.E. of regression	1.822188	Akaike info criterion	3.832650
Sum squared resid	8805.618	Schwarz criterion	3.841521
Log likelihood	-5080.010	Hannan-Quinn criter.	3.835861
Durbin-Watson stat	2.105627		

About 10% of yesterday's squared surprise (shock) carries into today's variance.

About 86% of yesterday's variance carries forward.

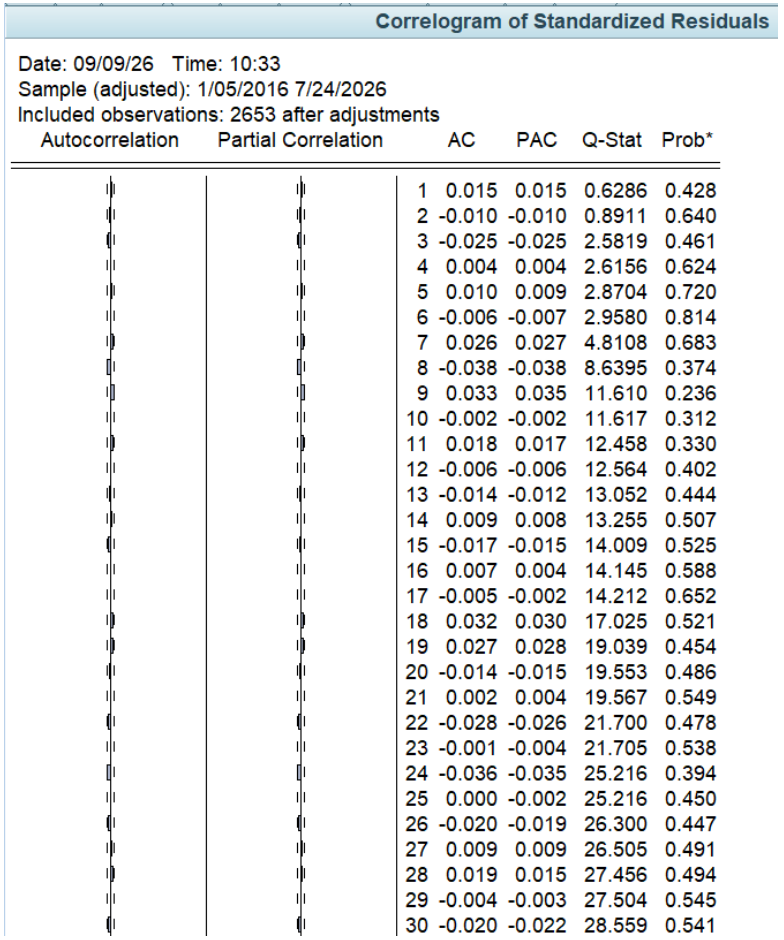
$\alpha_1 + \phi_1 = 0.0987 + 0.8562 = 0.955.$

Below 1 – the model is stationary and volatility is mean-reverting.

But it's high, meaning shocks to volatility die out only slowly.

EViews example.
GARCH(1,1) evaluation

Correlogram of standardized residuals



Correlogram of Standardized Residuals

Date: 09/09/26 Time: 10:33
Sample (adjusted): 1/05/2016 7/24/2026
Included observations: 2653 after adjustments

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob*
		1	0.015	0.015	0.6286	0.428
		2	-0.010	-0.010	0.8911	0.640
		3	-0.025	-0.025	2.5819	0.461
		4	0.004	0.004	2.6156	0.624
		5	0.010	0.009	2.8704	0.720
		6	-0.006	-0.007	2.9580	0.814
		7	0.026	0.027	4.8108	0.683
		8	-0.038	-0.038	8.6395	0.374
		9	0.033	0.035	11.610	0.236
		10	-0.002	-0.002	11.617	0.312
		11	0.018	0.017	12.458	0.330
		12	-0.006	-0.006	12.564	0.402
		13	-0.014	-0.012	13.052	0.444
		14	0.009	0.008	13.255	0.507
		15	-0.017	-0.015	14.009	0.525
		16	0.007	0.004	14.145	0.588
		17	-0.005	-0.002	14.212	0.652
		18	0.032	0.030	17.025	0.521
		19	0.027	0.028	19.039	0.454
		20	-0.014	-0.015	19.553	0.486
		21	0.002	0.004	19.567	0.549
		22	-0.028	-0.026	21.700	0.478
		23	-0.001	-0.004	21.705	0.538
		24	-0.036	-0.035	25.216	0.394
		25	0.000	-0.002	25.216	0.450
		26	-0.020	-0.019	26.300	0.447
		27	0.009	0.009	26.505	0.491
		28	0.019	0.015	27.456	0.494
		29	-0.004	-0.003	27.504	0.545
		30	-0.020	-0.022	28.559	0.541

ARCH LM test

Heteroskedasticity Test: ARCH

F-statistic	0.516416	Prob. F(10,2632)	0.8798
Obs*R-squared	5.175590	Prob. Chi-Square(10)	0.8791

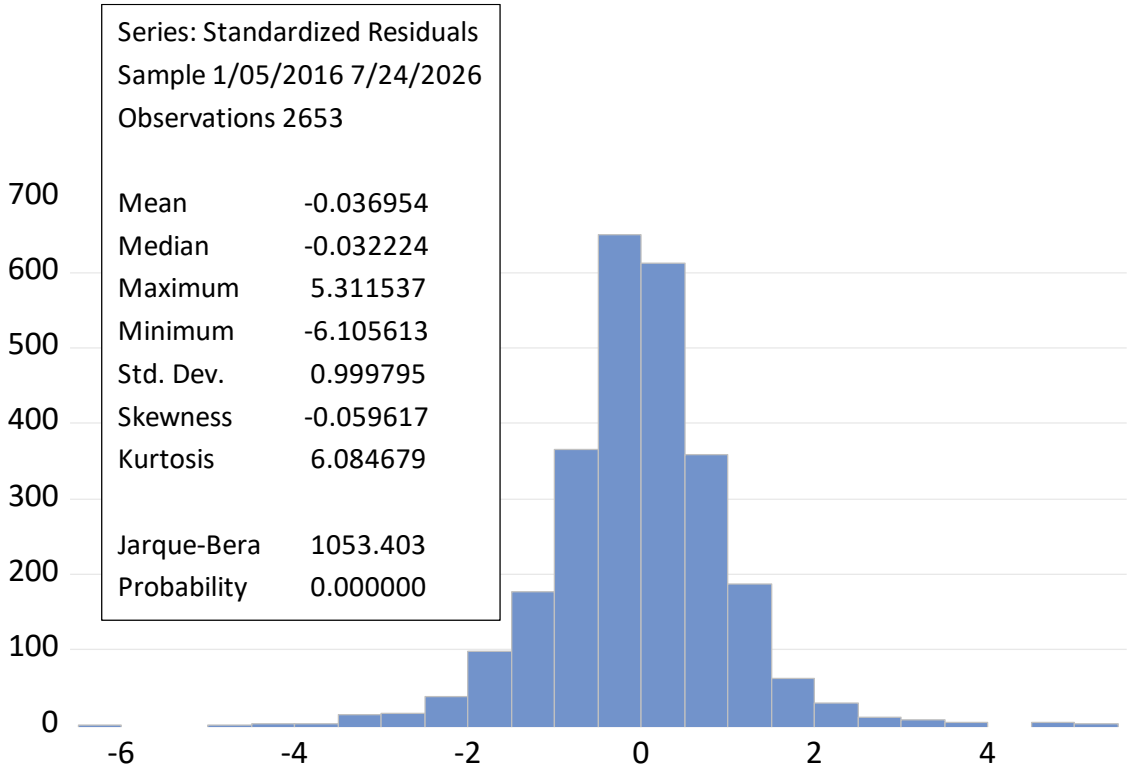
Test Equation:
Dependent Variable: WGT_RESID^2
Method: Least Squares
Date: 09/09/26 Time: 10:34
Sample (adjusted): 1/20/2016 7/24/2026
Included observations: 2643 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.031743	0.075709	13.62779	0.0000
WGT_RESID^2(-1)	0.006894	0.019492	0.353705	0.7236
WGT_RESID^2(-2)	-0.010238	0.019492	-0.525232	0.5995
WGT_RESID^2(-3)	-0.013979	0.019494	-0.717111	0.4734
WGT_RESID^2(-4)	0.024363	0.019491	1.249956	0.2114
WGT_RESID^2(-5)	0.002567	0.019497	0.131680	0.8952
WGT_RESID^2(-6)	0.005864	0.019497	0.300743	0.7636
WGT_RESID^2(-7)	-0.021271	0.019491	-1.091341	0.2752
WGT_RESID^2(-8)	0.002384	0.019500	0.122278	0.9027
WGT_RESID^2(-9)	-0.004824	0.019499	-0.247374	0.8046
WGT_RESID^2(-10)	-0.022536	0.019500	-1.155703	0.2479

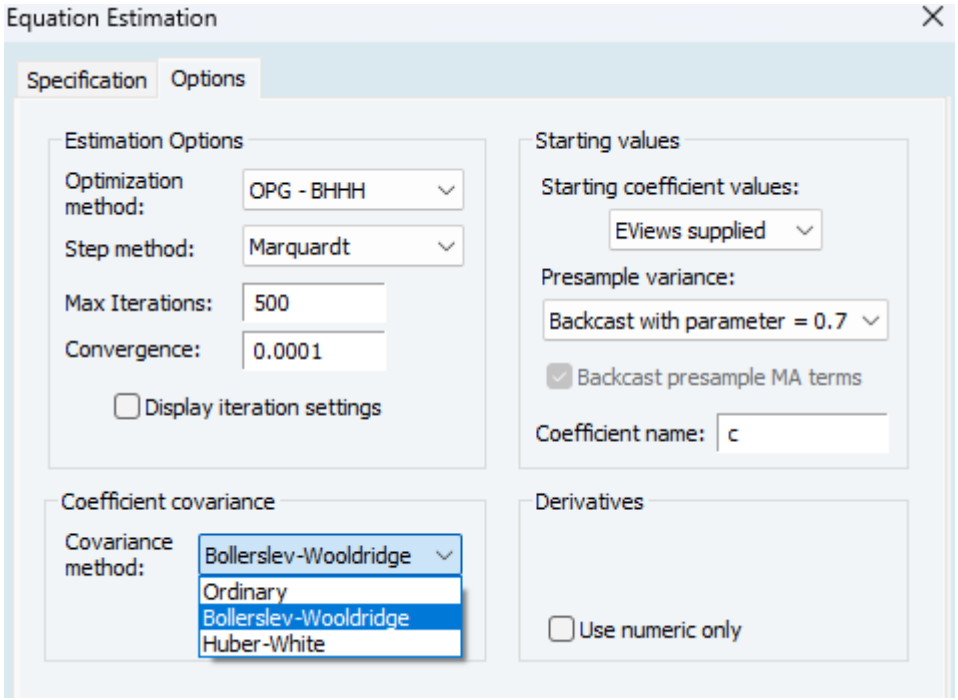
R-squared	0.001958	Mean dependent var	1.000983
Adjusted R-squared	-0.001834	S.D. dependent var	2.260357
S.E. of regression	2.262429	Akaike info criterion	4.474908
Sum squared resid	13472.11	Schwarz criterion	4.499379
Log likelihood	-5902.591	Hannan-Quinn criter.	4.483767
F-statistic	0.516416	Durbin-Watson stat	1.999894
Prob(F-statistic)	0.879751		

EViews example.
GARCH(1,1) evaluation: normality (1)

Residuals do not follow normal distribution



For describing volatility dynamics and doing inference we must use robust standard errors



EViews example.
GARCH(1,1) evaluation: normality (2)

Baseline

Dependent Variable: DLOG(PRICE)*100
Method: ML ARCH - Normal distribution (OPG - BHHH / Marquardt steps)
Date: 09/09/26 Time: 10:31
Sample (adjusted): 1/05/2016 7/24/2026
Included observations: 2653 after adjustments
Convergence achieved after 38 iterations
Coefficient covariance computed using outer product of gradients
Presample variance: backcast (parameter = 0.7)
GARCH = C(2) + C(3)*RESID(-1)^2 + C(4)*GARCH(-1)

Variable	Coefficient	Std. Error	z-Statistic	Prob.
C	0.157004	0.028397	5.528819	0.0000
Variance Equation				
C	0.142350	0.016576	8.587796	0.0000
RESID(-1)^2	0.098731	0.009773	10.10281	0.0000
GARCH(-1)	0.856236	0.012623	67.83337	0.0000

R-squared	-0.001136	Mean dependent var	0.095632
Adjusted R-squared	-0.001136	S.D. dependent var	1.821154
S.E. of regression	1.822188	Akaike info criterion	3.832650
Sum squared resid	8805.618	Schwarz criterion	3.841521
Log likelihood	-5080.010	Hannan-Quinn criter.	3.835861
Durbin-Watson stat	2.105627		

Robust standard errors

Dependent Variable: DLOG(PRICE)*100
Method: ML ARCH - Normal distribution (OPG - BHHH / Marquardt steps)
Date: 09/09/26 Time: 10:36
Sample (adjusted): 1/05/2016 7/24/2026
Included observations: 2653 after adjustments
Convergence achieved after 38 iterations
Coefficient covariance computed using Bollerslev-Wooldridge QML sandwich with expected Hessian
Presample variance: backcast (parameter = 0.7)
GARCH = C(2) + C(3)*RESID(-1)^2 + C(4)*GARCH(-1)

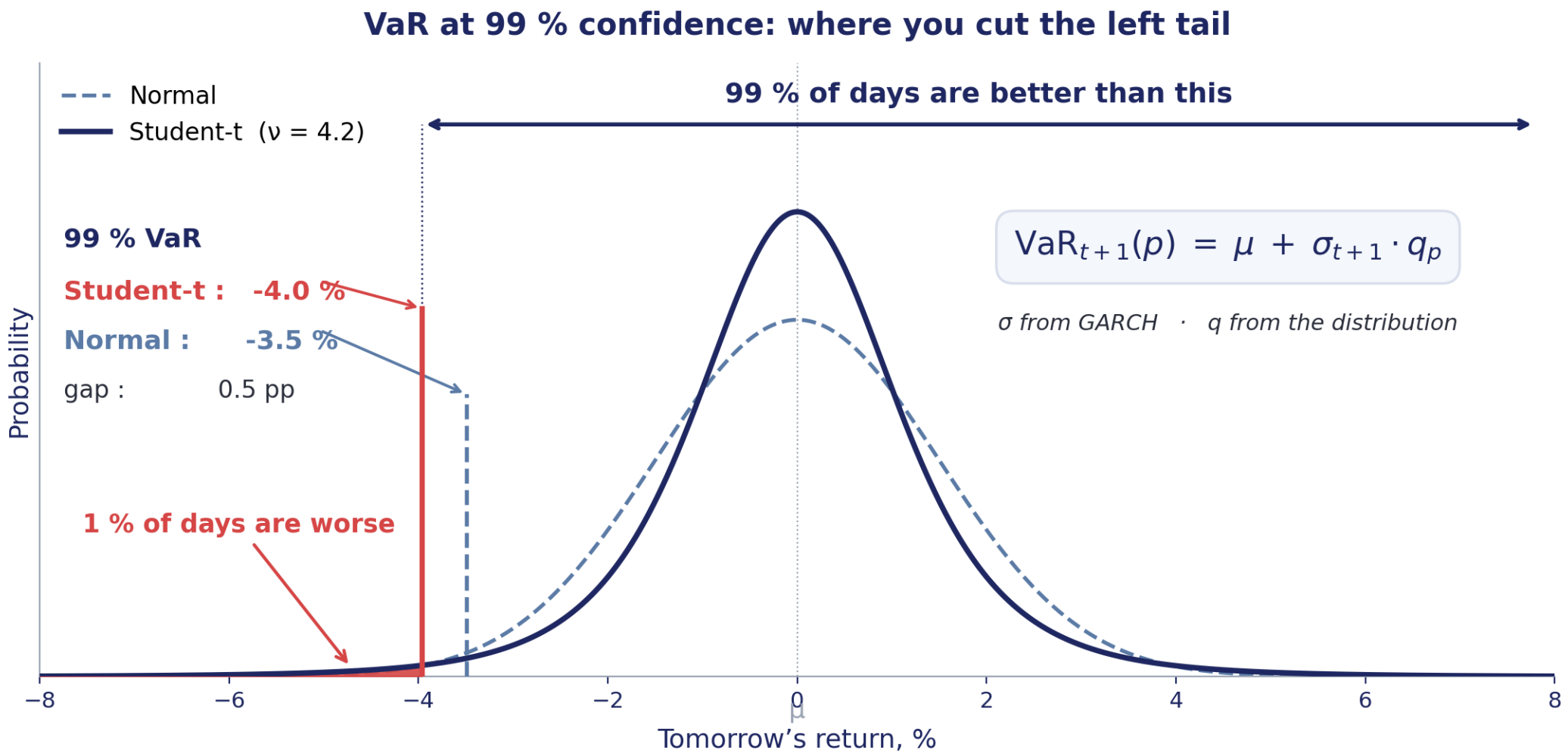
Variable	Coefficient	Std. Error	z-Statistic	Prob.
C	0.157004	0.030874	5.085350	0.0000
Variance Equation				
C	0.142350	0.042088	3.382186	0.0007
RESID(-1)^2	0.098731	0.015949	6.190359	0.0000
GARCH(-1)	0.856236	0.023325	36.70848	0.0000

R-squared	-0.001136	Mean dependent var	0.095632
Adjusted R-squared	-0.001136	S.D. dependent var	1.821154
S.E. of regression	1.822188	Akaike info criterion	3.832650
Sum squared resid	8805.618	Schwarz criterion	3.841521
Log likelihood	-5080.010	Hannan-Quinn criter.	3.835861
Durbin-Watson stat	2.105627		

If you're going to compute VaR (Value-at-Risk) robust standard errors don't fix the non-normality issue.

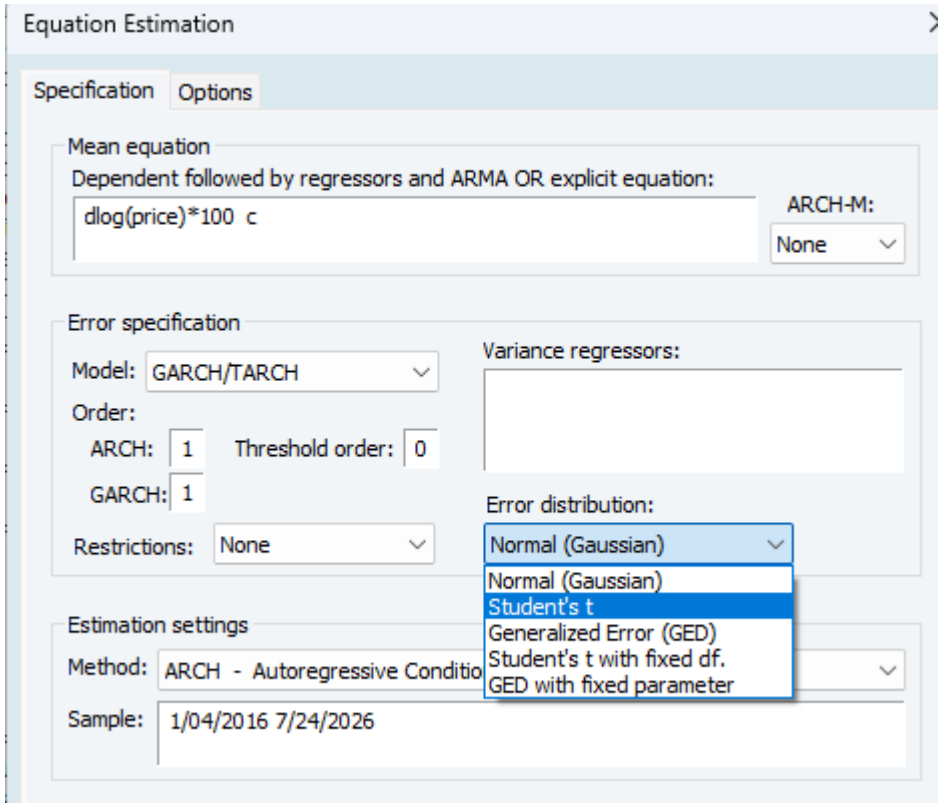
To account for the non-normality we can assume *t*-distribution of errors instead of normal.

Value-at-Risk (VaR)



EViews example. GARCH(1,1): *t* distribution

We assume Student's *t* distribution of the errors



Adjusted model

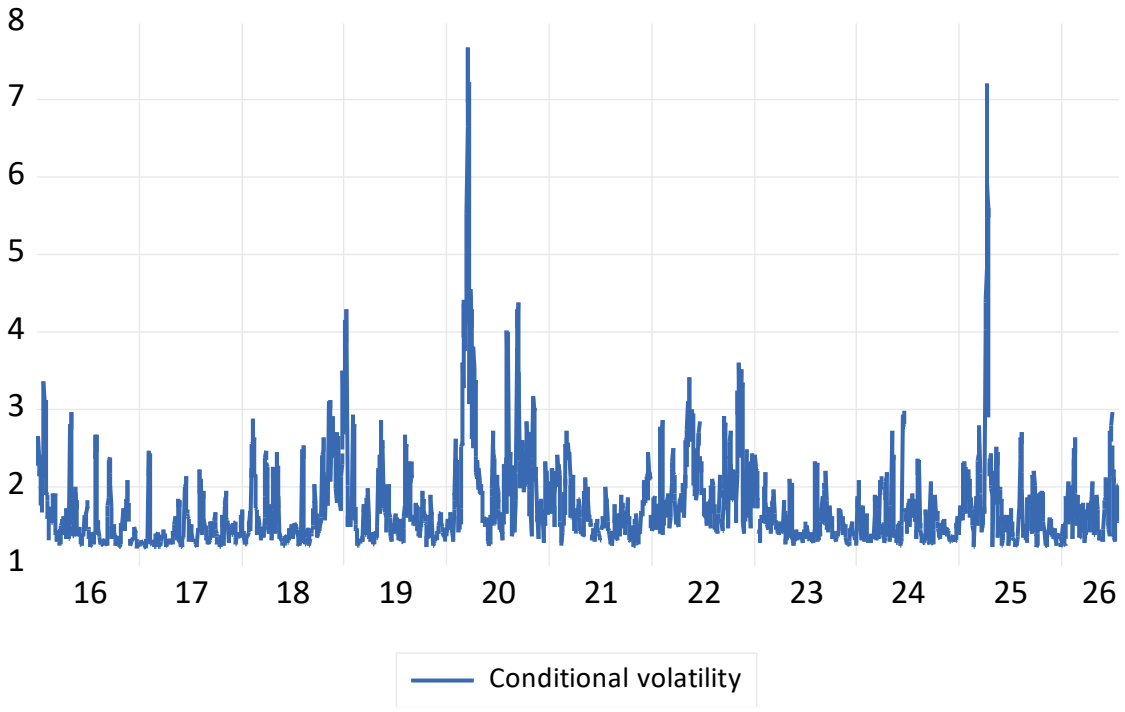
Dependent Variable: DLOG(PRICE)*100
Method: ML ARCH - Student's t distribution (OPG - BHHH / Marquardt steps)
Date: 09/09/26 Time: 10:38
Sample (adjusted): 1/05/2016 7/24/2026
Included observations: 2653 after adjustments
Convergence achieved after 20 iterations
Coefficient covariance computed using outer product of gradients
Presample variance: backcast (parameter = 0.7)
GARCH = C(2) + C(3)*RESID(-1)^2 + C(4)*GARCH(-1)

Variable	Coefficient	Std. Error	z-Statistic	Prob.
C	0.135081	0.025722	5.251584	0.0000
Variance Equation				
C	0.099210	0.025314	3.919215	0.0001
RESID(-1)^2	0.107316	0.017224	6.230662	0.0000
GARCH(-1)	0.871244	0.018197	47.87929	0.0000
T-DIST. DOF	4.235289	0.392031	10.80345	0.0000

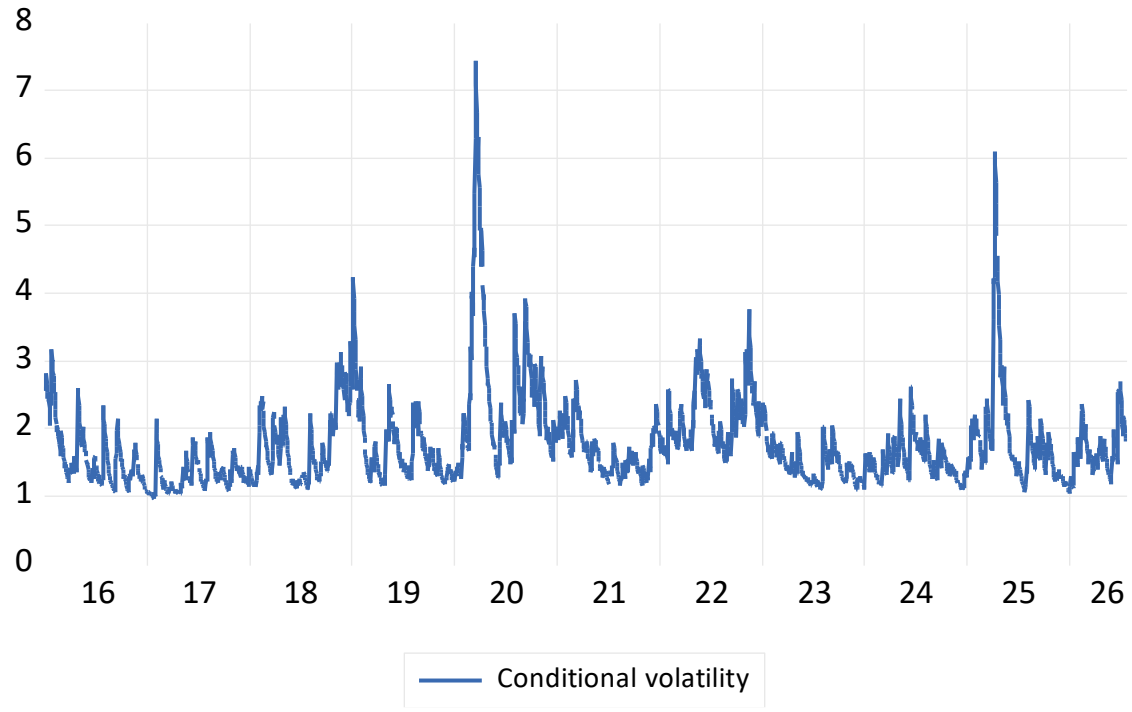
R-squared	-0.000469	Mean dependent var	0.095632
Adjusted R-squared	-0.000469	S.D. dependent var	1.821154
S.E. of regression	1.821581	Akaike info criterion	3.729106
Sum squared resid	8799.754	Schwarz criterion	3.740194
Log likelihood	-4941.659	Hannan-Quinn criter.	3.733119
Durbin-Watson stat	2.107031		

Conditional volatility. ARCH(5) vs GARCH(1,1)

ARCH(5)



GARCH(1,1)



4. EGARCH Model

Problem:
GARCH treats good and bad news alike

$$\sigma_t^2 = \alpha_0 + \phi_1 \sigma_{t-1}^2 + \alpha_1 u_{t-1}^2$$

The shock enters **squared** – the model sees only its **magnitude**, never its **sign**.

▲ **+5% day**

good news

=

▼ **–5% day**

bad news

Identical effect on tomorrow's volatility

But financial markets don't behave this way – a price drop raises volatility far more than an equal rise

Volatility asymmetry: Leverage effect

1	**Price falls**	A drop in the stock price raises the firm’s debt-to-equity ratio
2	**Equity gets riskier**	Higher leverage makes the remaining equity more volatile
3	**Volatility rises**	Bad news therefore feeds volatility more strongly than good news does

Beyond accounting: risk premia, panic selling and market psychology all push the same way

Testing for asymmetric effects (1)

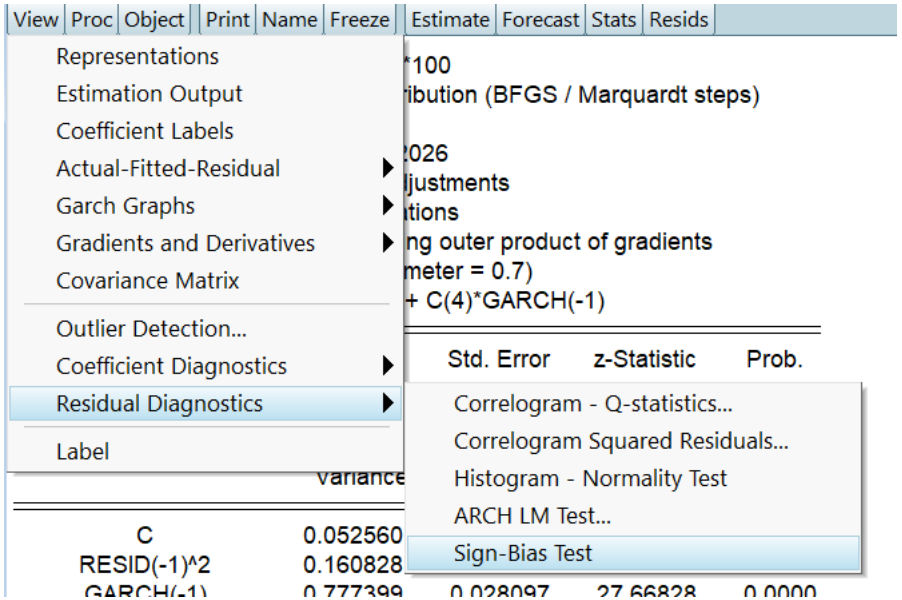
GARCH(1,1) model for FTSE-100

Dependent Variable: DLOG(PRICE)*100
Method: ML ARCH - Student's t distribution (BFGS / Marquardt steps)
Date: 09/09/26 Time: 10:45
Sample (adjusted): 1/05/2016 7/24/2026
Included observations: 2667 after adjustments
Convergence achieved after 27 iterations
Coefficient covariance computed using outer product of gradients
Presample variance: backcast (parameter = 0.7)
GARCH = C(2) + C(3)*RESID(-1)^2 + C(4)*GARCH(-1)

Variable	Coefficient	Std. Error	z-Statistic	Prob.
C	0.052661	0.013017	4.045552	0.0001
Variance Equation				
C	0.052560	0.010603	4.957209	0.0000
RESID(-1)^2	0.160828	0.022974	7.000508	0.0000
GARCH(-1)	0.777399	0.028097	27.66828	0.0000
T-DIST. DOF	5.191060	0.542637	9.566356	0.0000

R-squared	-0.001098	Mean dependent var	0.021052
Adjusted R-squared	-0.001098	S.D. dependent var	0.954154
S.E. of regression	0.954678	Akaike info criterion	2.331314
Sum squared resid	2429.818	Schwarz criterion	2.342354
Log likelihood	-3103.807	Hannan-Quinn criter.	2.335309
Durbin-Watson stat	2.022046		

Engle-Ng sign-bias test



Testing for asymmetric effects (2)

Engle-Ng test results

Engle-Ng Sign-Bias Test
Null Hypothesis: No leverage effects in standardized residuals

	t-Statistic	Prob.
Sign-Bias	0.587167	0.5571
Negative-Bias	-2.256306	0.0241
Positive-Bias	-1.364419	0.1726
Joint-Bias	16.49480	0.0009

Test Equation
Dependent Variable: RESID^2
Method: Least Squares
Date: 09/09/26 Time: 10:47
Sample (adjusted): 1/06/2016 7/24/2026
Included observations: 2666 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.938952	0.080887	11.60824	0.0000
SMINUS(-1)	0.066036	0.112465	0.587167	0.5571
SMINUS(-1)*RESID(-1)	-0.169387	0.075073	-2.256306	0.0241
SPLUS(-1)*RESID(-1)	-0.127934	0.093764	-1.364419	0.1726

R-squared	0.006158	Mean dependent var	0.989415
Adjusted R-squared	0.005038	S.D. dependent var	2.154541
S.E. of regression	2.149106	Akaike info criterion	4.369481
Sum squared resid	12294.87	Schwarz criterion	4.378315
Log likelihood	-5820.518	Hannan-Quinn criter.	4.372677
F-statistic	5.498267	Durbin-Watson stat	1.992641
Prob(F-statistic)	0.000918		

Reading

The effect lives in the **negative-size bias**: larger **drops** drive volatility up, positive shocks of equal size do not.

Joint test rejects symmetry **($p < 0.001$)**.

Conclusion

A symmetric GARCH is misspecified – we need a model that lets the sign of the shock matter

Exponential GARCH (EGARCH) model

$$\log(\sigma_t^2) = \alpha_0 + \alpha_1|z_{t-1}| + \gamma z_{t-1} + \phi_1 \log(\sigma_{t-1}^2)$$

γ Sign term → asymmetry

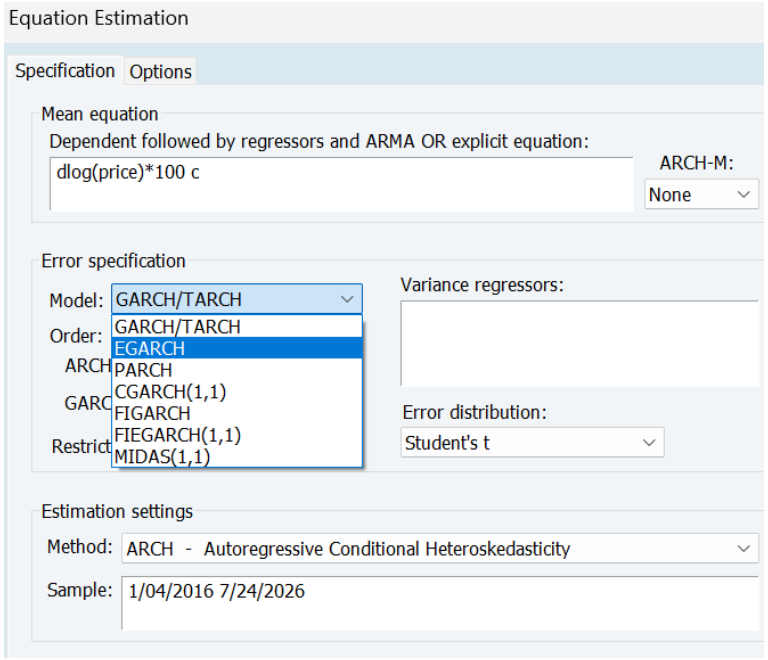
The extra term $\gamma \cdot z_{t-1}$ carries the **sign** of the shock, so good and bad news can differ. A **negative γ** means bad news raises volatility more.

log Log-variance → no constraints

Modelling $\log(\sigma^2)$ guarantees $\sigma^2 = \exp(\cdot) > 0$ automatically – so **no non-negativity constraints** are needed. Coefficients may take any sign.

$z_{t-1} = u_{t-1} / \sigma_{t-1}$ is the standardized shock

EGARCH(1,1) in EViews



Log-variance intercept

Size (magnitude) effect

Asymmetry / leverage effect

Log-variance persistence

EGARCH(1,1) model for FTSE-100

Dependent Variable: DLOG(PRICE)*100
Method: ML ARCH - Student's t distribution (BFGS / Marquardt steps)
Date: 09/09/26 Time: 10:48
Sample (adjusted): 1/05/2016 7/24/2026
Included observations: 2667 after adjustments
Convergence achieved after 36 iterations
Coefficient covariance computed using outer product of gradients
Presample variance: backcast (parameter = 0.7)
LOG(GARCH) = C(2) + C(3)*ABS(RESID(-1)/@SQRT(GARCH(-1))) + C(4)*RESID(-1)/@SQRT(GARCH(-1)) + C(5)*LOG(GARCH(-1))

Variable	Coefficient	Std. Error	z-Statistic	Prob.
C	0.029268	0.012821	2.282928	0.0224
Variance Equation				
C(2)	-0.151186	0.019368	-7.805996	0.0000
C(3)	0.172502	0.024062	7.169141	0.0000
C(4)	-0.142814	0.016632	-8.586433	0.0000
C(5)	0.959260	0.007093	135.2444	0.0000
T-DIST. DOF	5.760253	0.633766	9.088931	0.0000

R-squared	-0.000074	Mean dependent var	0.021052
Adjusted R-squared	-0.000074	S.D. dependent var	0.954154
S.E. of regression	0.954189	Akaike info criterion	2.311849
Sum squared resid	2427.333	Schwarz criterion	2.325097
Log likelihood	-3076.850	Hannan-Quinn criter.	2.316643
Durbin-Watson stat	2.024116		

Impact on log-variance

Positive shock: $\alpha_1 + \gamma = = 0.1725 - 0.1428 = \mathbf{0.030}$

Negative shock: $\alpha_1 - \gamma = = 0.1725 - (-0.1428) = \mathbf{0.315}$

A price drop raises volatility about 10x as much as an equal rise

Alternative specifications

- GARCH-M: allows the conditional mean to depend on its own conditional variance.
- Adding explanatory variables in the mean equation (e.g. UIP).
- Adding explanatory variables in the variance equation.
- etc.

Summary.
ARCH/GARCH models estimation procedure (1)

Stage 1 Identifying the effects of conditional heteroscedasticity

– The best model for the time series is constructed in the ARIMA model class or in the regression model class. Residuals for the estimated model are calculated.

– The ARCH LM test is performed to detect conditional heteroscedasticity.

Summary.
ARCH/GARCH models estimation procedure (2)

Stage 2 Estimating a model with conditional heteroscedasticity (ARCH/GARCH)

– Check the covariance-stationarity of the mean and variance equations.

– In the variance equation an intercept must be positive, ARCH and GARCH coefficients must be non-negative.

– Evaluate the model adequacy using the standard tools and statistics. Adjust the model if necessary.

– Test for the leverage effects. Estimate EGARCH if they are present.